Transformer-Based **Short-form Video Captioning & Analytics** using *Multi-Scale Attention*

- Vishwani Singh, Caleb Lee, Ricky Lee, Harsh Mamania

What data looks like, dataset description

SoccerNet-Caption: Has 36,894 temporally anchored rich textual commentaries describing 715.9 hours of soccer games (471 SoccerNet untrimmed broadcast games)

- Each caption describes a single significant event (5-10 secs), not every second is captioned
- Each annotation has game time, caption and label

[TEAM] try to speed play up by taking a quick free kick.

[PLAYER] gets a well deserved ovation! [PLAYER] ([TEAM]) produces a decent first...

What input looks like

{'**features**': tensor([[-0.3967, -0.4567, 0.0034, ..., -0.1578, -1.0014, 0.2074],

[-0.4039, -0.4447, -0.0355, ..., -0.0441, -0.7590, 0.3366],

[-0.0224, -0.3208, -0.0372, ..., -0.5116, -0.4866, 0.1246],

...,

[-0.3674, -0.4351, -0.3059, ..., 0.0214, -0.4052, 0.6806],

[-0.2236, -0.1819, -0.2424, ..., 0.0232, -0.5544, 0.5278],

[-0.0776, -0.5284, -0.3748, ..., -0.0975, -0.8826, 0.2551]]),

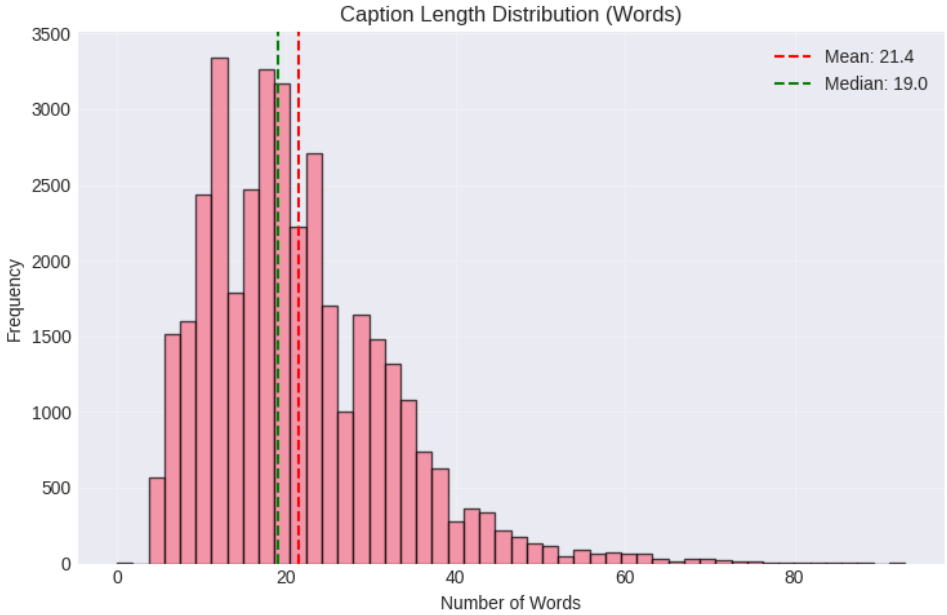
'**caption**': '[PLAYER] ([TEAM]) volleys the ball from just outside the box, but his effort goes narrowly over the bar.',

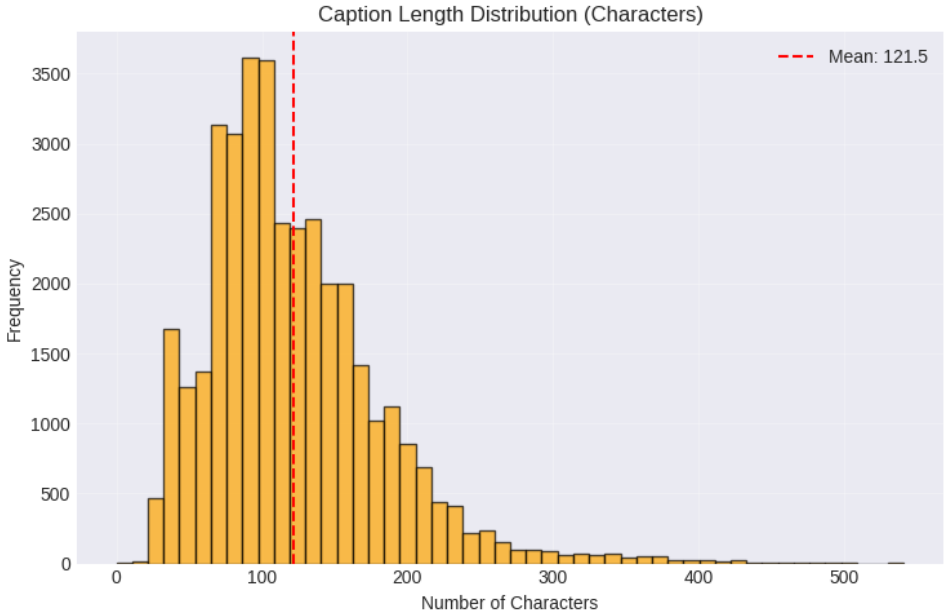
'**match_id**': 'england_epl/2014-2015/2015-02-21 - 18-00 Chelsea 1 - 1 Burnley',

'**timestamp**': '2621000',

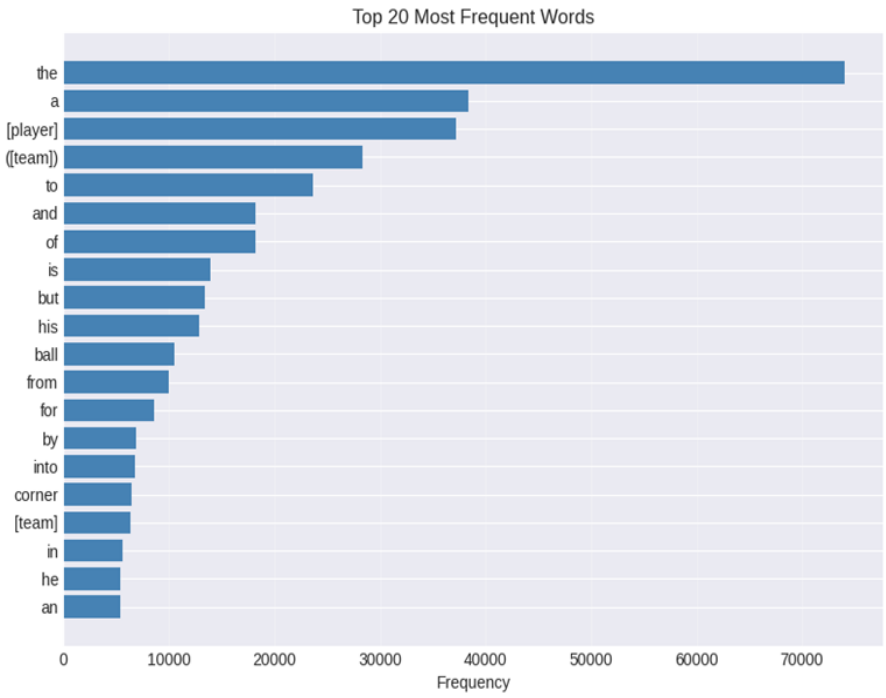
'**tokens**': [1, 5, 6, 3, 4, 14, 16, 109, 161,.....]

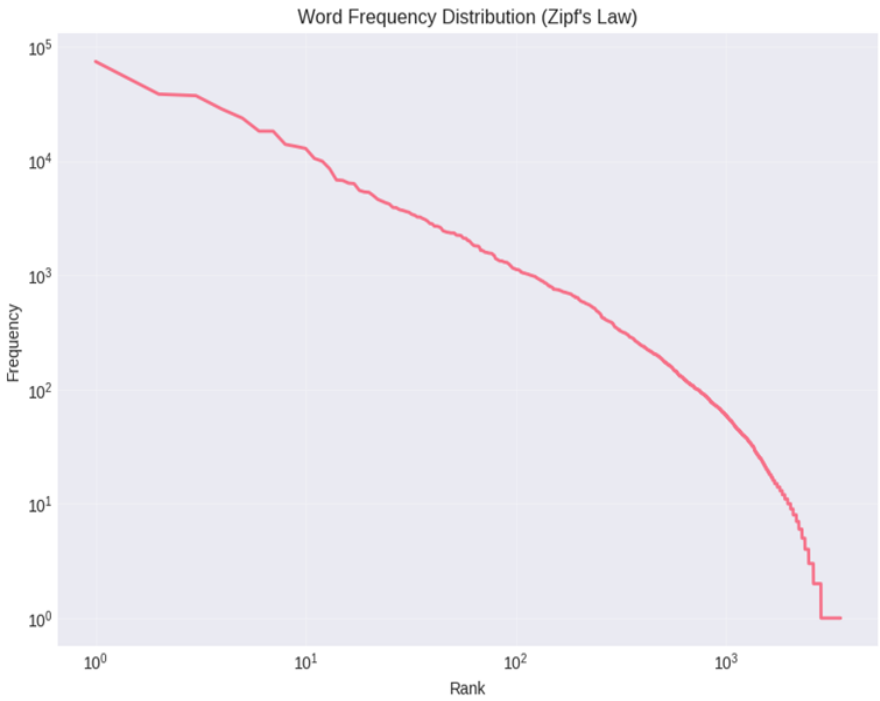
EDA


Caption Length Distribution (Words)
Mean: 21.4
Median: 19.0
Frequency
Number of Words
0
500
1000
1500
2000
2500
3000
3500
0
20
40
60
80

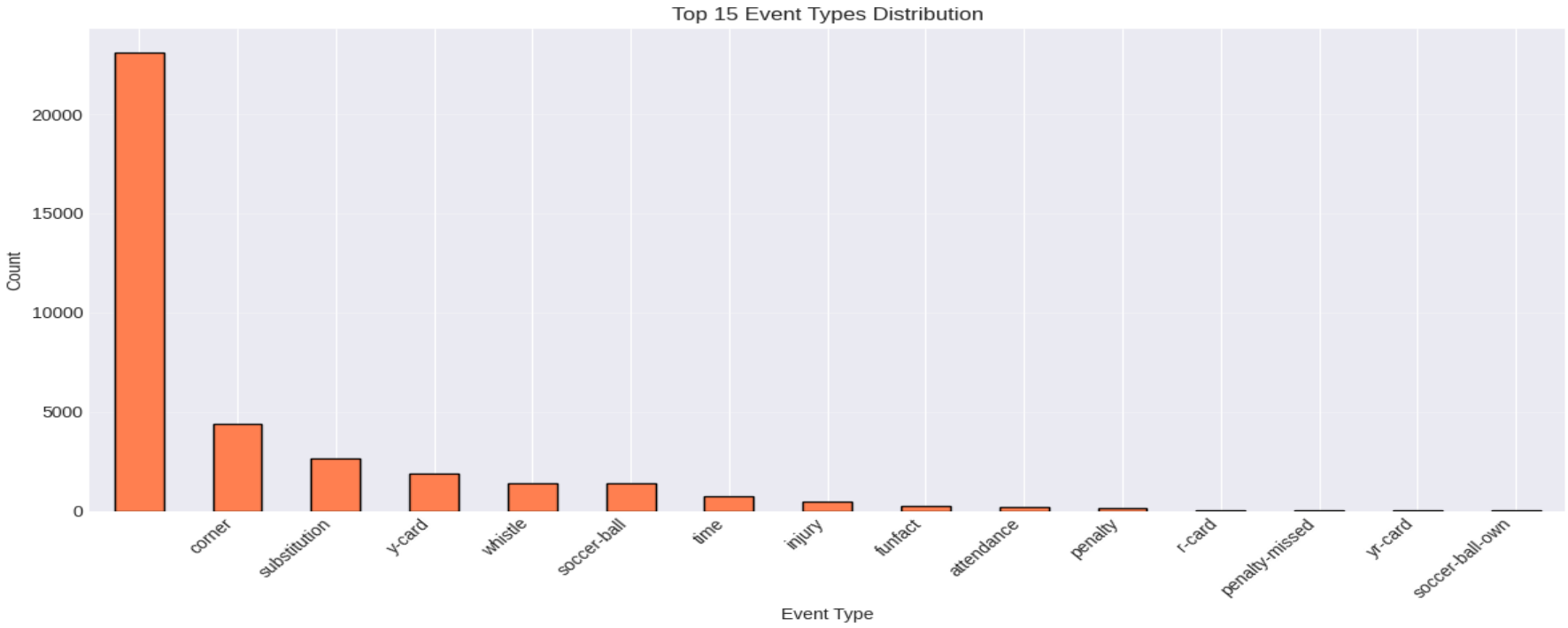

Caption Length Distribution (Characters)
Mean: 121.5
Frequency
Number of Characters
0
500
1000
1500
2000
2500
3000
3500
0
100
200
300
400
500

Frequently Used Words





Event Distribution Type



Video Captioning

Pipeline

```
STAGE 1: DATA ACQUISITION

SoccerNet-Caption Dataset
• 2,713 training samples
• 471 validation samples
• 168 test samples
• Each sample: video clip + ground truth caption

STAGE 2: FRAME EXTRACTION

• Extract 16 evenly-spaced frames per video
• Resize to 224×224 (ViT input size)
• Normalize with ImageNet mean/std
• Output: [16, 3, 224, 224] tensor per video

STAGE 3: FEATURE EXTRACTION (ViT)

• Model: google/vit-base-patch16-224 (pretrained, frozen)
• Process each frame independently
• Extract [CLS] token embedding per frame
• Output: [16, 768] features per video
• Features cached to disk (no recomputation)

STAGE 4: VOCABULARY BUILDING

• Tokenize all captions (NLTK word_tokenize)
• Filter by minimum frequency (min_freq=2)
• Special tokens: <PAD>, <SOS>, <EOS>, <UNK>
• Final vocabulary: 1,130 words

STAGE 5: CAPTION DECODER (Transformer)

Video Features [16, 768]
        ↓
Feature Projection (768 → 512)
        ↓
+ Frame Positional Encoding [16, 512]
        ↓
   TRANSFORMER DECODER (×6)
   • Multi-Head Attention (8)   ←── Token Embeddings + Pos Enc
   • Cross-Attention to Frames
   • Feed-Forward Network
   • Layer Normalization
        ↓
Output Projection (512 → vocab_size)
        ↓
Beam Search Decoding (k=3)
        ↓
Generated Caption

STAGE 6: OUTPUT

"player team sends a cross into the box but player comes off his
 line to gather the ball"
```

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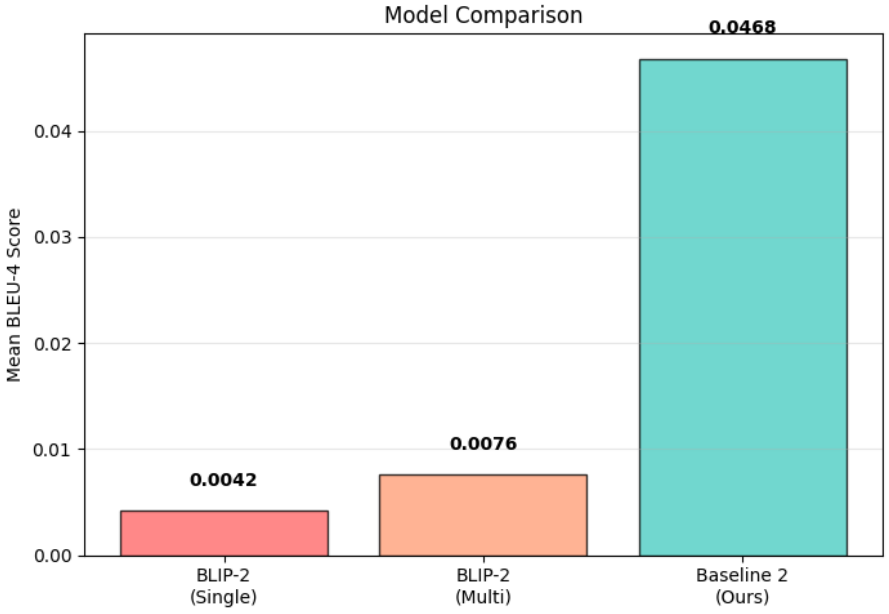
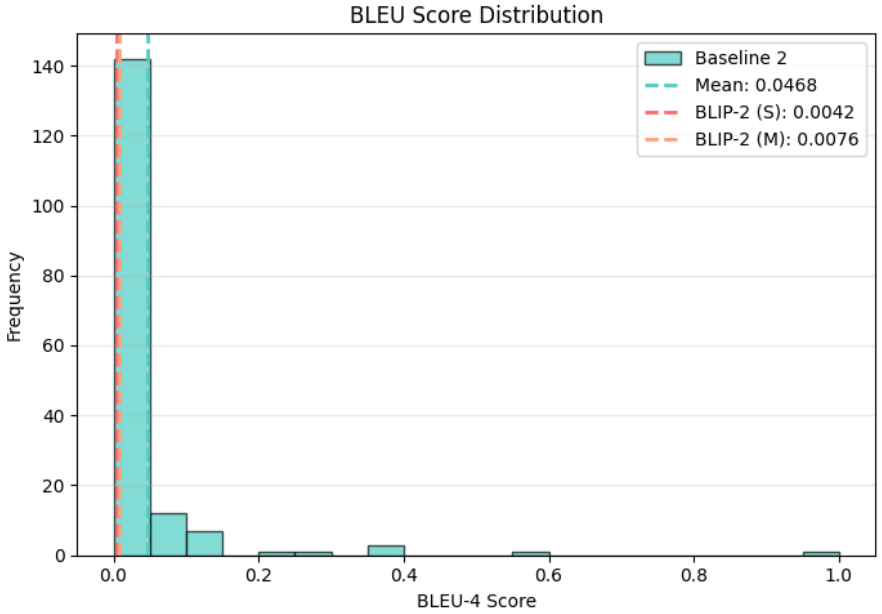
Baseline - Architecture

```
BASELINE ARCHITECTURE

Video Features    [16 frames × 768 dim]
      ↓
Feature Proj      768 → 512
      ↓
   TRANSFORMER DECODER
  • Layers: 4
  • Attention Heads: 1          ← Single head = limited
  • d_model: 512                  pattern learning
  • d_ff: 2048
  • Dropout: 0.4 (applied 4×)   ← Too much regularization

  ✗ No frame positional encoding ← No temporal awareness
      ↓
Output Linear     512 → 961 (vocab)
      ↓
Greedy Decoding   ← Suboptimal sequence selection
```

```
CONFIGURATION

Vocabulary: 961 words (min_freq=3)
Max Caption Length: 30 tokens
Training Epochs: 5
Loss: Cross-Entropy only

RESULTS

BLEU-4: 0.0202
Vocabulary Used: 17 unique words

KEY ISSUES

1. Mode Collapse: Repeated same caption template 145+ times
2. Single Head: Cannot learn multiple attention patterns
3. No Temporal Encoding: Treats frames as unordered bag
4. High Dropout: Underfitting on small dataset
5. Greedy Decoding: Locally optimal but globally suboptimal
```

Baseline - Result (BLEU score)





Top 5 Results

BLEU: 1.0000
GT: player team sends a cross into the box but player comes off his line to gather the ball
Gen: player team sends a cross into the box but player comes off his line to gather the ball

BLEU: 0.5954
GT: the corner from player team is cleared away by the defence
Gen: the corner from player team is intercepted by the defence

BLEU: 0.3742
GT: player team sends a chip pass into the box but player is quickest to intercept the ball
Gen: player team sends a cross into the box but player comes off his line to gather the ball

BLEU: 0.3602
GT: a pass from player team is intercepted by player and the move breaks down
Gen: the corner from player team is intercepted by the defence

BLEU: 0.3564
GT: the resulting corner from player team is cleared by the opposition 's defence
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Bottom 5 Results

BLEU: 0.0000
GT: 5 additional min will be played
Gen: the resulting corner from player team is cleared

BLEU: 0.0000
GT: fourth official shows 1 min of added time
Gen: the resulting corner from player team is cleared

BLEU: 0.0000
GT: we will have 3 min of added time
Gen: the resulting corner from player team is cleared by the defence

BLEU: 0.0000
GT: fourth official shows 2 min of added time
Gen: the resulting corner from player team is cleared by the defence

BLEU: 0.0000
GT: 5 min of stoppage-time to be played
Gen: the resulting corner from player team is cleared

Modifications:

IMPROVEMENT 1: MULTI-HEAD ATTENTION

Change: 1 head → 8 heads

Why: Different heads learn different patterns

- Head 1: Focus on player positions
- Head 2: Focus on ball trajectory
- Head 3: Focus on early frames (setup)
- Head 4: Focus on late frames (result)

Impact: Eliminated mode collapse, 70% more diverse vocabulary

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Change: 4 layers → 6 layers

Why: More capacity for complex sequence patterns

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- Middle layers: Phrase structure learning
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Impact: Better phrase-level accuracy (BLEU-4 +132%)

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Change: No frame positions → Learnable frame embeddings

Why: Model now understands temporal sequence

- Frame 1-4: Setup/context
- Frame 5-12: Main action
- Frame 13-16: Result/outcome

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IMPROVEMENT 4: REDUCED DROPOUT
Change: 0.4 dropout (4× application) → 0.1 dropout (2× application)
Why: Small dataset (2,713 samples) was underfitting

- Old: Too much regularization → couldn't learn patterns
- New: Lighter regularization → better pattern capture

Impact: Lower perplexity (64.4 → 4.0), more confident predictions

IMPROVEMENT 5: EXPANDED VOCABULARY

Change: min_freq=3 (961 words) → min_freq=2 (1,130 words)

Why: Fewer <UNK> tokens during training

- +169 words added to vocabulary
- Includes: "intercepted", "defence", "clearance", etc.

Impact: Model can generate more diverse, specific captions

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IMPROVEMENT 6: BEAM SEARCH DECODING

How it works:

- Maintain top-3 candidate sequences at each step
- Score = log_prob - 0.5 × repetition_penalty
- Select globally optimal sequence

Impact: Better sequence-level decisions, no repetition loops

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IMPROVEMENT 7: CUSTOM LOSS FUNCTION

Change: Cross-Entropy only → Multi-component loss

Formula: **L = CE + 0.1×Repetition + 0.1×Coverage - 0.05×Diversity**

Components:

- **CE Loss**: Standard next-token prediction
- **Repetition**: Penalize probability mass on recent tokens
- **Coverage**: Ensure all 16 frames receive attention
- **Diversity**: Encourage different heads to attend differently

Impact: Balanced attention, no repetition, diverse outputs

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IMPROVEMENT 8: EXTENDED TRAINING

Change: 5 epochs → 50 epochs (with early stopping patience=5)

Result: Best model at Epoch 12 (BLEU-4: 0.077 validation)

Impact: Proper convergence, not undertrained

Change in Results:

BLEU Score Statistics:

- Mean BLEU-4: 0.0468
- Median BLEU-4: 0.0240
- Std Dev: 0.1019
- Min: 0.0000
- Max: 1.0000

📊 BLEU-4: 0.0468
📊 CIDEr: 0.2036
📊 METEOR: 0.1222
📊 ROUGE-L: 0.2353
📊 Perplexity: 4.01

Metric	Old Model	New Model	Improvement
BLEU-1	0.1311	0.1997	**+52.3%**
BLEU-2	0.0563	0.1262	**+124.1%**
BLEU-3	0.0302	0.0702	**+132.6%**
BLEU-4	0.0202	0.0472	**+133.7%**
METEOR	0.1232	0.2056	**+66.9%**
ROUGE-L	0.2829	0.2427	**+-14.2%**

Best & Worst

✓ Best 5:

Sample 134: BLEU 1.0000
GT: player team sends a cross into the box but player comes off his line to gather the ball
Gen: player team sends a cross into the box but player comes off his line to gather the ball

Sample 33: BLEU 0.5954
GT: the corner from player team is cleared away by the defence
Gen: the corner from player team is intercepted by the defence

Sample 151: BLEU 0.3742
GT: player team sends a chip pass into the box but player is quickest to intercept the ball
Gen: player team sends a cross into the box but player comes off his line to gather the ball

Sample 48: BLEU 0.3602
GT: a pass from player team is intercepted by player and the move breaks down
Gen: the corner from player team is intercepted by the defence

Sample 135: BLEU 0.3564
GT: the resulting corner from player team is cleared by the opposition 's defence
Gen: the corner from player team is intercepted by the defence

✓ Worst 5:

Sample 0: BLEU 0.0000
GT: 5 additional min will be played
Gen: the resulting corner from player team is cleared

Sample 55: BLEU 0.0000
GT: fourth official shows 1 min of added time
Gen: the resulting corner from player team is cleared

Sample 89: BLEU 0.0000
GT: we will have 3 min of added time
Gen: the resulting corner from player team is cleared by the defence

Sample 130: BLEU 0.0000
GT: fourth official shows 2 min of added time
Gen: the resulting corner from player team is cleared by the defence

Sample 156: BLEU 0.0000
GT: 5 min of stoppage-time to be played
Gen: the resulting corner from player team is cleared

Net Improvement

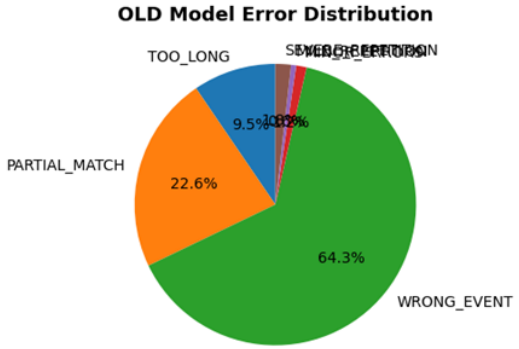
```
DETAILED IMPROVEMENT STATISTICS (BLEU-4)
============================================================

📊 SAMPLE COUNTS
   Improved:    123 ( 73.2%)
   Worse:        40 ( 23.8%)
   Same:          5 (  3.0%)

📈 IMPROVEMENTS (n=123)
   Average gain:  +0.0426
   Median gain:   +0.0139
   Max gain:      +0.7859
   Min gain:      +0.0002

📉 REGRESSIONS (n=40)
   Average drop:  -0.0177
   Median drop:   -0.0039
   Max drop:      -0.2075
   Min drop:      -0.0001

🎯 OVERALL
   Mean change:   +0.0270
   Median change: +0.0087
   Std dev:       0.0922

⚖️ NET IMPACT
   Total gained:  +5.2447
   Total lost:    -0.7083
   Net gain:      +4.5365
   Gain/Loss ratio: 7.41x

📊 IMPROVEMENT MAGNITUDE BREAKDOWN
   Small  (0-0.05):    105 samples
   Medium (0.05-0.15):  11 samples
   Large  (0.15-0.30):   2 samples
   Huge   (>0.30):       5 samples

📊 REGRESSION MAGNITUDE BREAKDOWN
   Small  (0-0.05):     36 samples
   Medium (0.05-0.15):   3 samples
   Large  (>0.15):       1 samples
```

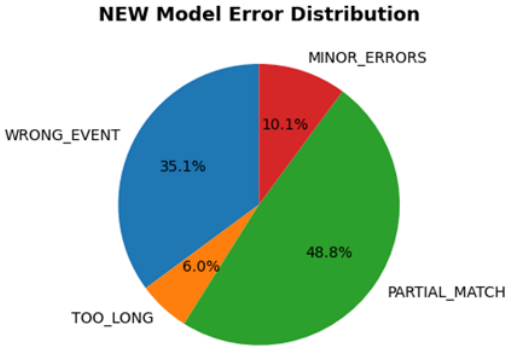
Error Type Analysis

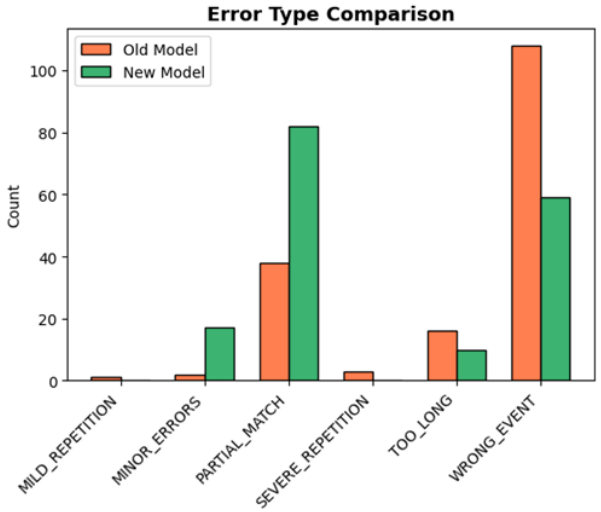
ERROR TYPE	OLD MODEL	NEW MODEL	CHANGE
MILD_REPETITION	1 (0.6%)	0 (0.0%)	-1
MINOR_ERRORS	2 (1.2%)	17 (10.1%)	+15
PARTIAL_MATCH	38 (22.6%)	82 (48.8%)	+44
SEVERE_REPETITION	3 (1.8%)	0 (0.0%)	-3
TOO_LONG	16 (9.5%)	10 (6.0%)	-6
WRONG_EVENT	108 (64.3%)	59 (35.1%)	-49

AVERAGE ERROR METRICS COMPARISON

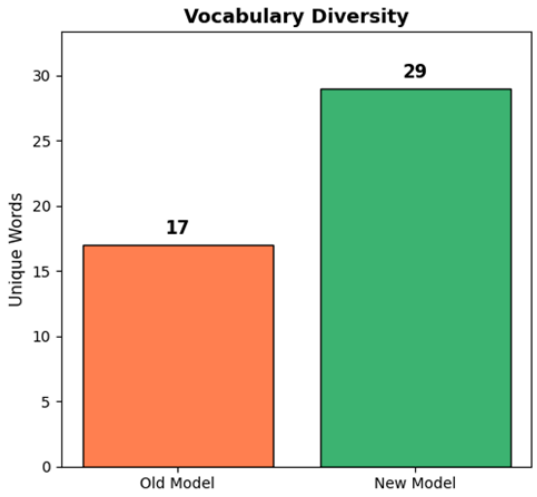
METRIC	OLD MODEL	NEW MODEL
missing_words	14.83	12.77
extra_words	8.95	6.46
repeated_bigrams	0.07	0.00
unk_count	0.00	0.00
common_words	2.74	3.93

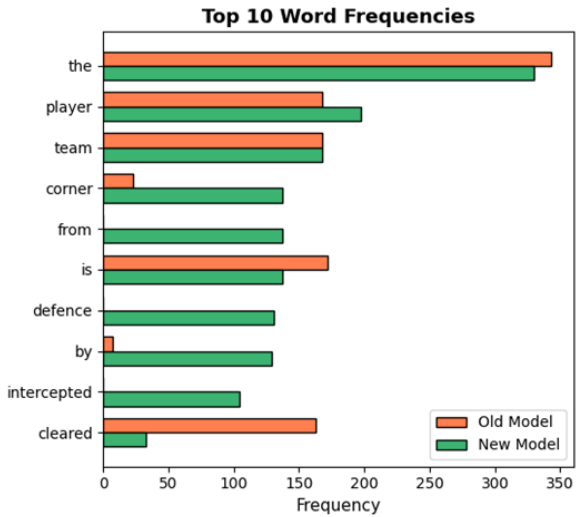


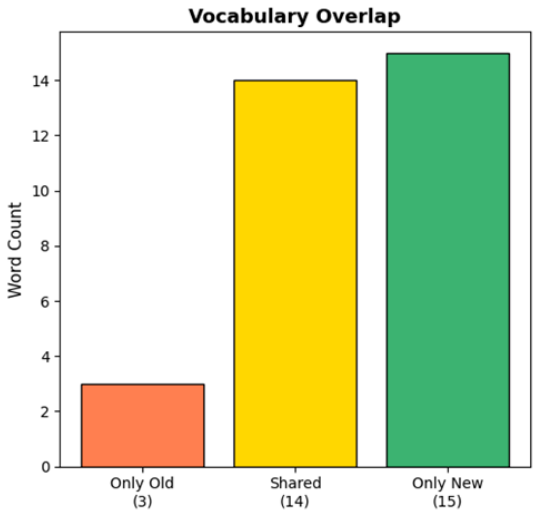




Vocabulary usage comparison -







📚 UNIQUE WORDS USED
Old Model: 17 unique words
New Model: 29 unique words
Difference: +12

📊 TOTAL WORDS GENERATED
Old Model: 2154 words
New Model: 1937 words

🔄 VOCABULARY OVERLAP
Common to both: 14 words
Only in Old: 3 words
Only in New: 15 words

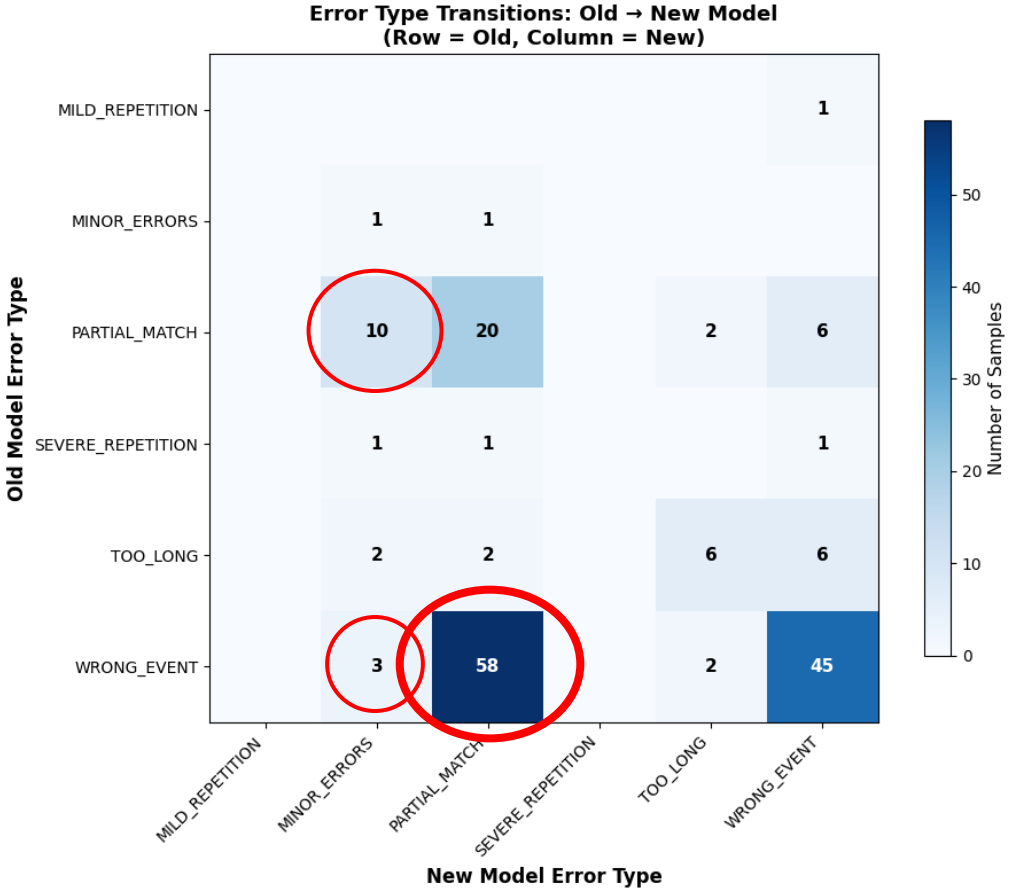
🏆 TOP 15 WORDS - OLD MODEL

the	343x
ball	175x
is	172x
player	168x
team	168x
but	168x
cleared	163x
sends	145x
a	145x
cross	145x
into	145x
box	145x
takes	23x
corner	23x
kick	10x

🏆 TOP 15 WORDS - NEW MODEL

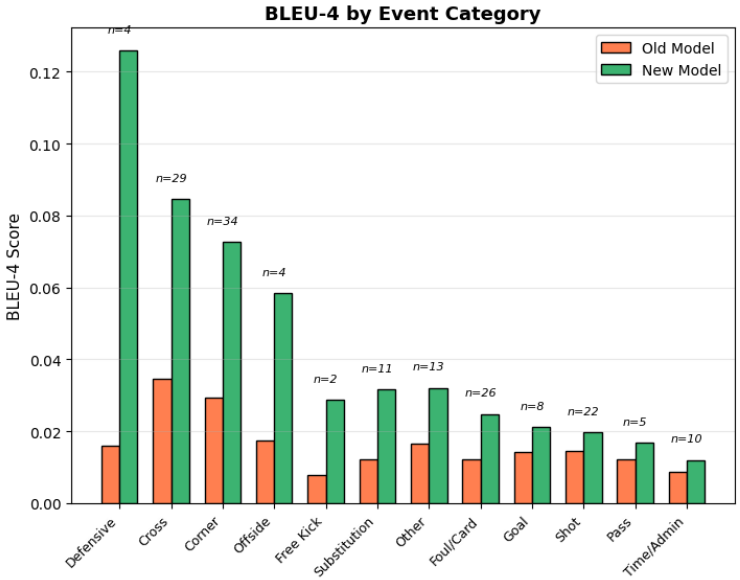
the	330x
player	197x
team	168x
corner	137x
from	137x
is	137x
defence	131x
by	129x
intercepted	104x
cleared	33x
sends	31x
a	31x
cross	31x
into	31x
box	31x

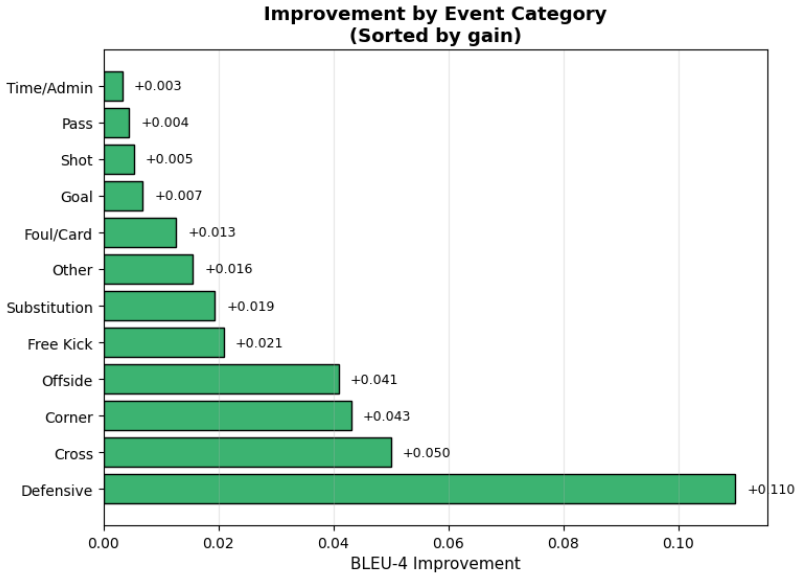
Confusion Matrix of Error Types (Old → New Transition)




```
================================================
KEY ERROR TRANSITIONS
================================================
✅ IMPROVEMENTS (moved to better category):
   WRONG_EVENT → PARTIAL_MATCH: 58 samples
   WRONG_EVENT → MINOR_ERRORS: 3 samples
   PARTIAL_MATCH → MINOR_ERRORS: 10 samples
   SEVERE_REPETITION → PARTIAL_MATCH: 1 samples
   SEVERE_REPETITION → MINOR_ERRORS: 1 samples
   TOO_LONG → PARTIAL_MATCH: 2 samples
   TOO_LONG → MINOR_ERRORS: 2 samples

⚠️ REGRESSIONS (moved to worse category):
   MINOR_ERRORS → PARTIAL_MATCH: 1 samples
   PARTIAL_MATCH → WRONG_EVENT: 6 samples
   PARTIAL_MATCH → TOO_LONG: 2 samples

🔄 STAYED SAME:
   MINOR_ERRORS: 1 samples unchanged
   PARTIAL_MATCH: 20 samples unchanged
   TOO_LONG: 6 samples unchanged
   WRONG_EVENT: 45 samples unchanged
```

Improvement by GT Category - Which event types improved most?



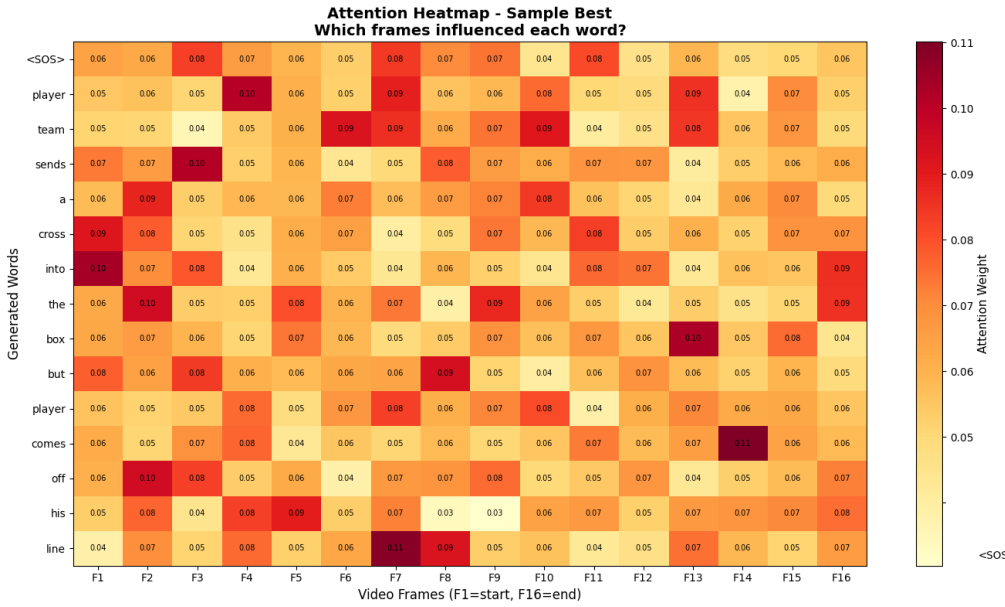


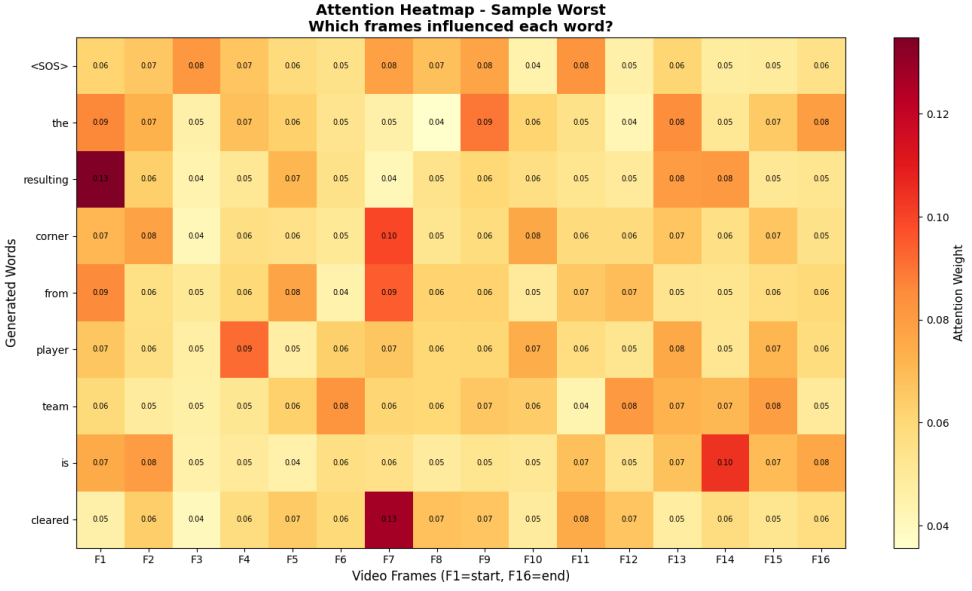

```
======================================================================
IMPROVEMENT BY EVENT CATEGORY
======================================================================

CATEGORY          COUNT   OLD BLEU   NEW BLEU    IMPROVE
----------------------------------------------------------------------
Corner               34     0.0294     0.0725    +0.0432
Cross                29     0.0345     0.0845    +0.0500
Foul/Card            26     0.0122     0.0248    +0.0126
Shot                 22     0.0144     0.0197    +0.0053
Other                13     0.0165     0.0321    +0.0156
Substitution         11     0.0123     0.0317    +0.0194
Time/Admin           10     0.0086     0.0120    +0.0033
Goal                  8     0.0142     0.0211    +0.0069
Pass                  5     0.0123     0.0168    +0.0045
Offside               4     0.0175     0.0585    +0.0410
Defensive             4     0.0160     0.1259    +0.1099
Free Kick             2     0.0078     0.0287    +0.0210
```

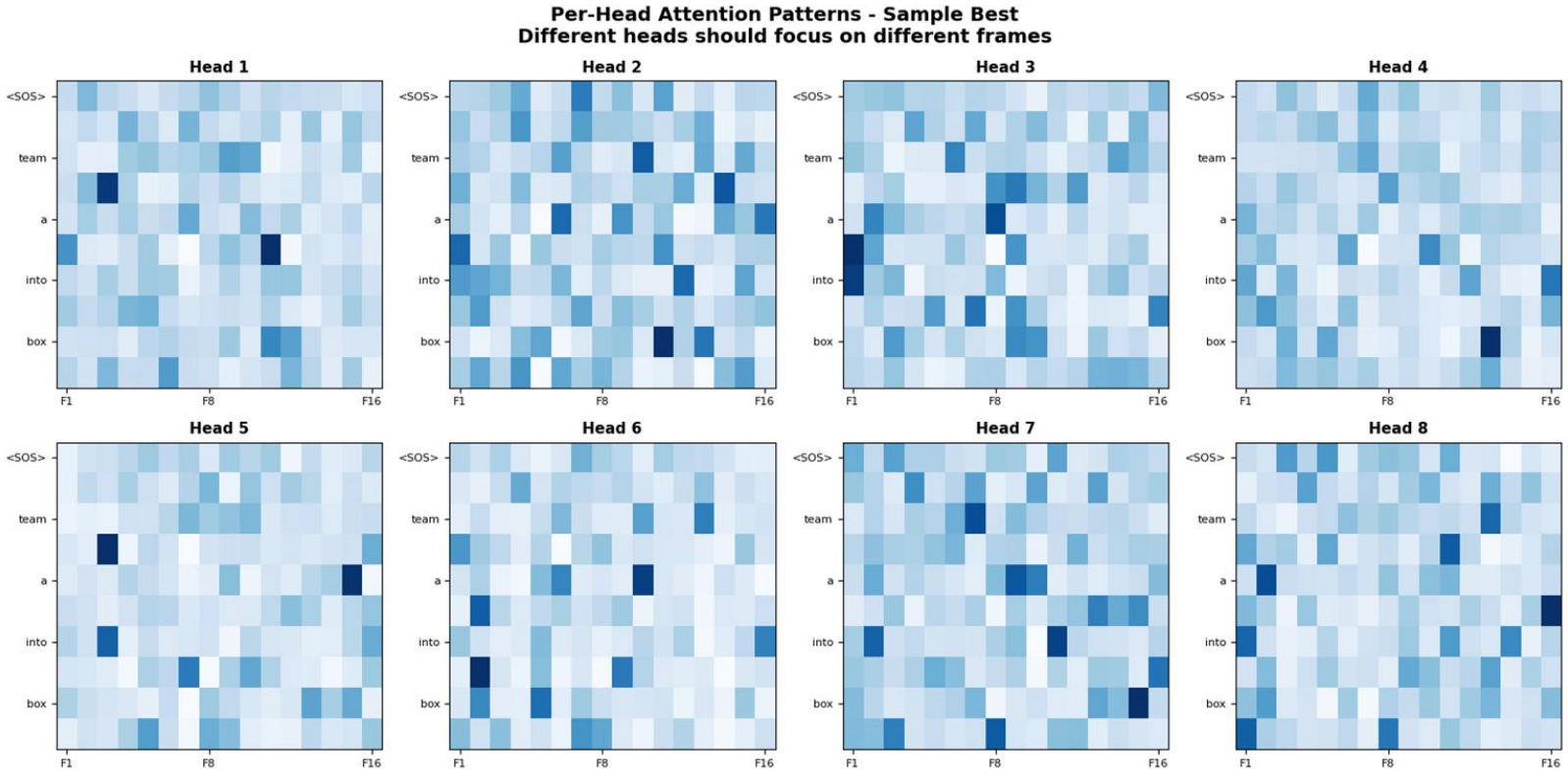
ATTENTION VISUALIZATION

--- BEST (Sample 134, BLEU: 1.0000) ---
GT: player team sends a cross into the box but player comes off his line to gather the ball
Gen: player team sends a cross into the box but player comes off his line to gather the ball

--- WORST (Sample 0, BLEU: 0.0000) ---
GT: 5 additional min will be played
Gen: the resulting corner from player team is cleared


Attention Heatmap - Sample Best
Which frames influenced each word?
Generated Words
<SOS>
player
team
sends
a
cross
into
the
box
but
player
comes
off
his
line
F1 F2 F3 F4 F5 F6 F7 F8 F9 F10 F11 F12 F13 F14 F15 F16
Video Frames (F1=start, F16=end)
Attention Weight
0.11
0.10
0.09
0.08
0.07
0.06
0.05


Attention Heatmap - Sample Worst
Which frames influenced each word?
Generated Words
<SOS>
the
resulting
corner
from
player
team
is
cleared
F1 F2 F3 F4 F5 F6 F7 F8 F9 F10 F11 F12 F13 F14 F15 F16
Video Frames (F1=start, F16=end)
Attention Weight
0.12
0.10
0.08
0.06
0.04


Per-Head Attention Patterns - Sample Best
Different heads should focus on different frames
Head 1
Head 2
Head 3
Head 4
Head 5
Head 6
Head 7
Head 8
<SOS>
team
a
into
box
F1
F8
F16

Where both Models failed-

📊 Found 57 samples where BOTH models failed (BLEU < 0.02)

📊 Found 4 samples where new model regressed significantly

```
--------------------------------------------------
BOTH MODELS FAILED - BY CATEGORY:
--------------------------------------------------
   Shot: 13 samples
   Foul/Card: 11 samples
   Corner: 8 samples
   Cross: 7 samples
   Time/Admin: 6 samples
   Goal: 5 samples
   Pass: 3 samples
   Other: 2 samples
   Substitution: 1 samples
   Free Kick: 1 samples
```

WHY IT FAILED:

1. **NON-VISUAL EVENT**
   - GT describes something not visible (time announcement, scoreboard)

2. **ATTENTION PATTERN**
   - Model attends uniformly to frames
   - No distinctive visual signal found
   - Defaults to common pattern

3. **MODEL LIMITATION**
   - Can only describe what it "sees"
   - Admin events need OCR/audio

SOLUTIONS:

- Add OCR for scoreboard
- Filter non-visual events
- Multi-modal input (audio)

DEMO TIME